A Multi-Modal Text Analytics Framework for eCommerce Customer Reviews: Integrating Emotion Detection, Sentiment Analysis, and Topic Modelling

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Abstract

Background/Motivation: Understanding customer sentiment and emotional responses from eCommerce reviews is critical for businesses seeking to improve product quality, marketing strategy, and customer retention. While prior work has separately addressed emotion detection, sentiment classification, and topic discovery, limited research has integrated all three modalities into a unified analytical framework applied to eCommerce review data.

Methods: This paper presents a multi-modal text analytics pipeline integrating transformer-based emotion detection (DistilRoBERTa), lexicon-based sentiment analysis (TextBlob), and unsupervised topic modelling via Latent Dirichlet Allocation (LDA). The framework was applied to a dataset of 23,485 women's clothing eCommerce reviews and requires no manual annotation.

Results: Joy dominated customer emotional responses across all product categories (57% of reviews), while sizing and fit issues constituted the most recurrent negative theme across multiple LDA topics. Cross-modal analysis confirmed convergent validity between emotional, sentiment, and topical signals: the sizing-related topic consistently aligned with elevated sadness classifications and negative sentiment polarity.

Conclusions: The proposed zero-annotation framework offers actionable business intelligence for marketing, product development, inventory management, and customer retention, demonstrating the practical value of combining complementary NLP techniques applied to unlabelled review corpora.

Keywords: *emotion classification; sentiment analysis; topic modelling; LDA; eCommerce; NLP; DistilRoBERTa; TextBlob; customer reviews; text analytics; business intelligence; affective computing*

1. Introduction

The rapid growth of eCommerce has generated unprecedented volumes of user-generated text data. Customer reviews represent a rich source of behavioural and experiential signals that businesses can exploit to improve products and services. The global eCommerce market is projected to reach $55.6 trillion by 2027 (Business Wire, 2022), and with approximately 2 billion people accessing eCommerce platforms globally, the scale of available review data continues to expand (Munna et al., 2020).

Customer reviews vary widely in length, tone, and linguistic complexity. Users may express nuanced emotional states such as satisfaction, disappointment, or surprise, requiring fine-grained classification that goes beyond simple positive or negative polarity (Poria et al., 2017). Recurring themes such as sizing inconsistency or fabric quality are often distributed across thousands of reviews, requiring topic modelling techniques to surface systematically (Barde and Bainwad, 2017).

Prior research has largely addressed these challenges in isolation. Sentiment analysis identifies polarity (Vanaja and Belwal, 2018; Taboada et al., 2011), emotion classification captures affective states (Acheampong et al., 2021), and topic modelling discovers latent themes (Blei et al., 2003). Studies combining sentiment analysis and topic modelling—notably in hotel and airline review domains (Calheiros et al., 2017; Kwon et al., 2021)—demonstrate that integrated approaches yield richer insight than individual methods alone. However, the combination of all three modalities into a unified business analytics framework remains underexplored in the eCommerce domain, particularly for unlabelled corpora.

This paper addresses this gap by presenting a multi-modal text analytics framework that simultaneously applies transformer-based emotion detection, lexicon-based sentiment analysis, and LDA topic modelling to a large dataset of eCommerce clothing reviews. The framework requires no manual annotation, making it directly applicable to real-world review corpora at scale. The contributions of this work are threefold:

- A unified analytical pipeline integrating emotion detection, sentiment analysis, and topic modelling for large-scale eCommerce review analysis, extending prior two-modality frameworks (Calheiros et al., 2017; Kwon et al., 2021) with emotion detection as a third analytical layer.
- An empirical investigation demonstrating how multiple NLP techniques capture complementary and convergent signals from customer-generated text, including cross-modal validation between emotion, sentiment, and topic outputs.
- Actionable business intelligence for marketing, product development, inventory management, and customer retention, derived from a zero-annotation pipeline applicable to any large review corpus.

The remainder of this paper is structured as follows. Section 2 reviews related work across sentiment analysis, emotion classification, and topic modelling. Section 3 describes the dataset. Section 4 details the methodology. Section 5 presents results, Section 6 discusses business implications, Section 7 discusses limitations and future work, and Section 8 concludes.

2. Related Work

2.1 Sentiment Analysis in eCommerce

Sentiment analysis of eCommerce reviews has been widely studied. Huang et al. (2023) reviewed current techniques and identified opportunities for deeper semantic understanding beyond polarity classification. Ji et al. (2018) demonstrated the value of sentiment analysis for identifying product quality issues in eCommerce settings using fuzzy decision support models. Vanaja and Belwal (2018) applied aspect-level sentiment analysis to eCommerce data, demonstrating that fine-grained aspect analysis reveals more actionable signals than document-level polarity. Drus and Khalid (2019) conducted a systematic literature review of sentiment analysis on social media content, providing a comprehensive taxonomy of approaches applicable to user-generated review corpora.

Despite the maturity of this field, most eCommerce sentiment studies remain confined to binary or ternary polarity classification, limiting their ability to capture the emotional nuance present in customer language. The present work addresses this by complementing sentiment scoring with fine-grained emotion detection.

2.2 Emotion Classification

Emotion classification introduces finer granularity than sentiment polarity, distinguishing affective states such as joy, sadness, fear, and disgust as defined by Ekman's foundational taxonomy of basic emotions (Ekman, 1992). Acheampong et al. (2021) provided a comprehensive review of transformer-based approaches for text emotion detection, demonstrating strong performance of BERT-family models across multiple benchmark datasets. The DistilRoBERTa model employed in this study offers classification performance competitive with larger transformer architectures at substantially reduced computational cost (Sanh et al., 2019), making it well suited for inference at scale without fine-tuning on labelled data.

While emotion classification has been applied to social media and clinical text, its application to eCommerce review data remains limited. This study contributes empirical evidence of its utility in the retail review domain.

2.3 Topic Modelling and Combined Approaches

Topic modelling via Latent Dirichlet Allocation (Blei et al., 2003) has been applied across multiple review domains. Kwon et al. (2021) combined LDA topic modelling with sentiment analysis for airline review data, demonstrating that thematic decomposition combined with sentiment scoring yields more actionable service insights than either technique alone. Calheiros et al. (2017) applied a similar combined approach to hotel review analysis, confirming that integration produces richer characterisations of customer experience.

The present work extends this established two-modality paradigm by incorporating transformer-based emotion detection as a third analytical layer, enabling simultaneous characterisation of what customers discuss (topics), how they evaluate products (sentiment), and how they feel (emotion). To the best of the author's knowledge, no prior study has combined all three modalities in a unified zero-annotation framework for eCommerce review analysis.

3. Dataset

The dataset consists of 23,485 women's clothing eCommerce reviews sourced from Kaggle (https://www.kaggle.com/datasets/nicapotato/womens-ecommerce-clothing-reviews). Each review is associated with a product category label spanning six departments: Tops, Dresses, Bottoms, Jackets, Intimate, and Trend. The dataset contains no pre-assigned emotion labels, making it representative of real-world unlabelled review corpora to which the proposed zero-annotation framework is designed to apply.

Review lengths vary considerably, from single-sentence ratings to detailed multi-paragraph descriptions of product fit, fabric quality, and appearance. Tops and Dresses represent the largest product categories by review volume, with Trend and Jackets representing the smallest. This distribution introduces natural class imbalance across product categories, which is preserved throughout the analysis to reflect realistic deployment conditions.

No personally identifiable reviewer information was used in this study. The dataset is publicly available under open licence and does not require ethical approval for secondary analysis.

4. Methodology

Figure 1 provides an overview of the end-to-end analytical pipeline. Each review passes through preprocessing, then through three parallel analytical modules (emotion detection, sentiment analysis, topic modelling), with outputs subsequently combined for cross-modal analysis.

4.1 Text Preprocessing

All reviews underwent a standardised preprocessing pipeline prior to analysis: HTML tag removal, URL stripping, lowercasing, punctuation removal, stopword filtering using NLTK, and tokenisation. WordNet lemmatisation was applied to normalise word forms. Reviews with null or empty text content were excluded from analysis, resulting in a final corpus of 23,485 usable reviews. TF-IDF vectorisation (Roelleke and Wang, 2008) was applied for feature extraction prior to topic modelling.

4.2 Emotion Detection via DistilRoBERTa

Emotion classification was performed using the pre-trained j-hartmann/emotion-english-distilroberta-base model (Hartmann et al., 2022), available via HuggingFace Transformers. This model is fine-tuned on six benchmark emotion datasets spanning multiple domains and assigns each input text one of seven emotion labels—joy, sadness, neutral, surprise, fear, disgust, and anger—corresponding to Ekman's taxonomy of basic emotions (Ekman, 1992).

The model was applied in inference mode without further domain fine-tuning. This zero-shot application eliminates the requirement for labelled training data and makes the approach directly applicable to unlabelled corpora at scale. Each review received a single dominant emotion label corresponding to the highest-confidence softmax output. The computational cost of inference was manageable on a standard GPU; approximate inference time for the full corpus was under two hours.

4.3 Sentiment Analysis via TextBlob

Sentiment polarity was computed using the TextBlob lexicon-based library. Each review was assigned a continuous polarity score in the range $[-1, 1]$ and subsequently classified as positive (polarity > 0), negative (polarity < 0), or neutral (polarity $= 0$). The lexicon-based approach, consistent with Taboada et al. (2011), provides a computationally efficient complement to the transformer-based emotion module and enables direct comparison of valence-level signals against fine-grained emotional outputs. Limitations of lexicon-based sentiment analysis, particularly its sensitivity to contextual negation and domain-specific language, are acknowledged and discussed in Section 7.

4.4 Topic Modelling via LDA

Latent Dirichlet Allocation (Blei et al., 2003) was applied to the full preprocessed review corpus using scikit-learn's LatentDirichletAllocation implementation. CountVectorizer was applied with a maximum document frequency of 0.95 and minimum document frequency of 2 to filter uninformative and extremely rare terms. The hyperparameters alpha and beta were left at their scikit-learn defaults (alpha = 1/k, beta = 0.1). The number of topics was set to five following qualitative assessment of topic coherence and interpretability across a range of k values (k = 3 to 10). Word clouds were generated for each topic to facilitate qualitative interpretation of thematic content. Future work should apply quantitative coherence metrics such as the C_v measure to support topic number selection in a more principled manner.

4.5 Cross-Modal Analysis

Following individual module analyses, cross-modal visualisations were constructed to examine the relationships between the three analytical outputs. A heatmap of topic distribution across sentiment categories quantified the alignment between thematic content and sentiment polarity. A stacked bar chart of topic distribution across emotion classes enabled integrated interpretation of emotional and thematic signals. This cross-modal triangulation approach follows the analytical framework of Kwon et al. (2021) and Calheiros et al. (2017), extended here to incorporate the emotional dimension as a third modality.

5. Results

5.1 Emotion Distribution

Joy was the dominant emotion across the full dataset, accounting for approximately 13,500 instances (57.5% of the corpus), substantially exceeding all other emotion categories combined. Sadness was the second most frequent emotion (approximately 3,500 instances; 14.9%), followed by neutral responses (approximately 2,700; 11.5%) and surprise (approximately 2,500; 10.6%). Fear, disgust, and anger occurred in substantially smaller quantities, each below 700 instances. The near-absence of anger (<40 instances; <0.2%) is notable and may reflect a positive selection bias inherent in voluntary review datasets, where strongly dissatisfied customers may be less likely to submit detailed written feedback, or may express dissatisfaction through other channels such as returns rather than reviews.

Across product categories, joy remained dominant in all six departments. However, the Tops and Dresses categories showed relatively elevated proportions of sadness and neutral emotions compared to other categories, a finding that aligns with the elevated negative sentiment observed in Section 5.2 and warrants further product-level investigation.

Emotion Class	Count (approx.)	Percentage (%)	Sentiment Category	Count (approx.)	Percentage (%)
Joy	13,500	57.5	Positive	~21,000	~89.4
Sadness	3,500	14.9	Negative	~1,300	~5.5
Neutral	2,700	11.5	Neutral	~1,200	~5.1
Surprise	2,500	10.6	—	—	—
Fear	650	2.8	—	—	—
Disgust	600	2.6	—	—	—
Anger	<40	<0.2	—	—	—

Table 1. Distribution of emotion classes and sentiment categories across the review dataset (N = 23,485).

5.2 Sentiment Analysis Results

Sentiment analysis confirmed the broadly positive nature of the review corpus. Positive reviews accounted for approximately 21,000 instances (~89% of the dataset), while negative and neutral reviews each numbered approximately 1,200–1,400 instances (~5–6% each). The strong skew towards positive sentiment is consistent with the emotion distribution findings, where joy dominated across all categories.

The Dresses category received both the highest volume of positive reviews and the highest absolute count of negative reviews, consistent with its proportional dominance in the dataset. The convergence between high joy classification and high positive sentiment rates across the dataset provides initial cross-modal validation of the two independent analytical components.

5.3 LDA Topic Modelling Results

Five distinct topics were extracted from the review corpus. Table 2 summarises each topic with its top keywords and inferred thematic label. Topics were labelled based on manual inspection of the top-20 keywords per topic by the author.

Topic	Inferred Theme	Top Keywords	Dominant Sentiment
1	Fit and Appearance (Jeans/Pants)	love, fit, great, pant, jean, wear, color, perfect, look, length	Positive
2	Sizing Issues (Shirts/Dresses)	fit, size, small, shirt, dress, little, like, large, arm, look	Mixed / Negative
3	Sizing and Return Behaviour	size, small, fit, ordered, like, wear, store, run, tried, medium	Mixed
4	Fabric and Aesthetic Quality (Dresses/Skirts)	dress, look, fabric, like, color, skirt, love, fit, beautiful, flattering	Strongly Positive
5	Comfort and Wearability (Sweaters/Dresses)	love, great, sweater, wear, color, dress, fit, comfortable, look, perfect	Strongly Positive

Table 2. LDA topic summary: inferred themes, representative keywords, and dominant sentiment orientation.

Sizing inconsistency emerges as the most persistent negative theme, spanning Topics 2 and 3. The co-occurrence of terms such as tried, ordered, small, and medium in Topic 3 provides lexical evidence that customers are ordering multiple sizes prior to purchase, indicative of inadequate size guidance at the point of sale. Topics 4 and 5 are strongly positive in character, with keywords such as beautiful, flattering, comfortable, and perfect, suggesting high customer satisfaction with fabric quality and wearability.

5.4 Cross-Modal Analysis

Cross-modal analysis examined the relationships between the three analytical outputs. The heatmap of topic distribution across sentiment categories confirmed that positive sentiment dominated all five topics. Topic 4 (Fabric and Aesthetic Quality) had the highest volume of positive reviews (4,815), consistent with its strongly positive keyword profile. Topic 2 (Sizing Issues) had the highest proportion of negative sentiment reviews (approximately 380 negative instances, approximately 14% of Topic 2), consistent with its thematic focus on sizing problems.

Cross-modal analysis of topic distribution across emotion classes revealed that Topics 4 and 5 were most strongly associated with joy classifications, while Topic 2 showed higher proportional alignment with sadness. The convergence between independently derived emotion, sentiment, and topic signals—where the same sizing-related topic consistently aligns with both negative sentiment and elevated sadness—strengthens confidence in the validity of each individual analytical component and demonstrates the value of cross-modal triangulation as a systematic validation strategy for zero-annotation NLP pipelines.

6. Business Implications

The integrated framework produces actionable insights directly applicable to core eCommerce business functions (Singh et al., 2022; Huang et al., 2023):

- Marketing and Customer Success: The high prevalence of joy-classified reviews provides strong material for testimonials and social proof campaigns. Topic 4 and Topic 5 keywords (beautiful, flattering, comfortable, perfect) represent authentic customer language suitable for advertising copy without further processing.
- Product Development: Topics 2 and 3 reveal a persistent sizing inconsistency problem, particularly for shirts and dresses. The lexical co-occurrence of tried, ordered, small, and medium in Topic 3 suggests customers are purchasing multiple sizes to identify correct fit. Improved size guidance, virtual fit tools, or standardised size charts would directly address this pattern and are likely to reduce return rates.
- Inventory Management: Products associated with high-joy, positive-sentiment topics (Topics 4 and 5) represent strong-demand items with high customer satisfaction. Inventory levels for these categories can be prioritised accordingly, while products generating negative emotional signals warrant quality review prior to restocking decisions.
- Customer Retention: Elevated sadness and negative sentiment signals in the Tops and Dresses categories represent potential churn indicators. Proactive customer service outreach to reviewers exhibiting these emotional signals may reduce attrition, consistent with findings by Yang et al. (2020) on emotion-driven retention strategies.

These implications demonstrate that a zero-annotation NLP pipeline can generate strategically differentiated outputs comparable to those requiring expensive manual labelling or bespoke annotation efforts.

7. Discussion

The results demonstrate that combining transformer-based emotion detection, lexicon-based sentiment analysis, and LDA topic modelling produces a richer characterisation of customer experience than any single method provides alone. The convergence of findings across independently operated methods—where the sizing-related topic consistently aligns with elevated sadness and negative sentiment—provides cross-modal triangulation that strengthens confidence in the reliability of each individual component. This convergent validity mirrors findings in prior two-modality studies (Calheiros et al., 2017; Kwon et al., 2021) while extending them to the three-modality case.

The framework's primary practical strength is its ability to decompose complex review corpora into interpretable layers: what customers feel (emotion), how positively or negatively they evaluate products (sentiment), and what specific themes they discuss (topics). This decomposition is particularly valuable for product managers and marketing teams who require specific, actionable signals rather than aggregate scores.

Two primary limitations should be acknowledged. First, the lexicon-based sentiment approach may misclassify context-dependent language, including irony, negation, and domain-specific expressions (Apte and Sri Khetwat, 2019). The high positive skew (89% positive) may partly reflect genuine satisfaction but may also reflect TextBlob's tendency to assign positive polarity to hedged or ambiguous language. Future work should compare TextBlob against a fine-tuned transformer sentiment classifier such as cardiffnlp/twitter-roberta-base-sentiment to assess the robustness of polarity estimates.

Second, the current framework assigns a single dominant emotion label per review, whereas customers may simultaneously express multiple emotional states within a single text. Extending the framework to support multi-label emotion classification would increase fidelity to real emotional expression (Poria et al., 2017). The LDA topic count was selected based on qualitative interpretability; future work should apply quantitative coherence metrics such as the C_v measure to support topic number selection in a more principled manner.

The framework was evaluated on a single domain (women's clothing). Generalisation to other eCommerce domains such as electronics, furniture, or food cannot be assumed and represents an important direction for future validation. The dataset's inherent positive selection bias—where satisfied customers may be over-represented among reviewers—should also be acknowledged when interpreting the strong dominance of joy and positive sentiment.

8. Conclusion

This paper presented a multi-modal text analytics framework integrating transformer-based emotion detection, lexicon-based sentiment analysis, and LDA topic modelling for eCommerce customer review analysis. Applied to 23,485 women's clothing reviews without manual annotation, the framework identified joy as the dominant customer emotion (57.5% of the corpus), revealed persistent sizing inconsistency as the primary negative theme, and demonstrated strong cross-modal convergence between emotional, sentiment, and topical signals.

The framework offers practical value for eCommerce businesses seeking data-driven insights from customer feedback at scale, without the cost and time requirements of manual annotation. The cross-modal triangulation approach provides a systematic validation strategy for zero-annotation NLP pipelines applicable beyond the retail domain.

Future work will extend the approach to multi-label emotion classification, evaluate transferability across retail domains including electronics and furniture, apply quantitative coherence evaluation for topic number selection, and explore time-series sentiment tracking to monitor product performance trajectories over time.

Code Availability

The implementation and experimental code used in this study are openly available at: https://github.com/hasaniftikhar07/multimodal-ecommerce-text-analytics

Conflict of Interest

The author declares no conflict of interest.

References

Acheampong, F.A., Nunoo-Mensah, H. and Chen, W. (2021) 'Transformer models for text-based emotion detection: a review of BERT-based approaches', Artificial Intelligence Review, 54(8), pp. 5789–5829.

Apte, P. and Sri Khetwat, S. (2019) 'Text-based emotion analysis: feature selection techniques and approaches', in Emerging Technologies in Data Mining and Information Security. Singapore: Springer, pp. 837–847.

Barde, B.V. and Bainwad, A.M. (2017) 'An overview of topic modeling methods and tools', in Proceedings of the 2017 International Conference on Intelligent Computing and Control Systems (ICICCS). IEEE, pp. 745–750.

Blei, D.M., Ng, A.Y. and Jordan, M.I. (2003) 'Latent Dirichlet allocation', Journal of Machine Learning Research, 3, pp. 993–1022.

Business Wire (2022) Global E-Commerce Market Growth, Opportunities and Forecasts 2022–2027. Available at: https://www.businesswire.com/news/home/20220222005665/en/ (Accessed: 5 September 2024).

Calheiros, A.C., Moro, S. and Rita, P. (2017) 'Sentiment classification of consumer-generated online reviews using topic modeling', Journal of Hospitality Marketing and Management, 26(7), pp. 675–693.

Drus, Z. and Khalid, H. (2019) 'Sentiment analysis in social media and its application: systematic literature review', Procedia Computer Science, 161, pp. 707–714.

Ekman, P. (1992) 'An argument for basic emotions', Cognition and Emotion, 6(3–4), pp. 169–200.

Hartmann, J., Heitmann, M., Siebert, C. and Schamp, C. (2022) 'More than a feeling: accuracy and application of sentiment analysis', International Journal of Research in Marketing, 40(1), pp. 75–87.

Huang, H., Zavareh, A.A. and Mustafa, M.B. (2023) 'Sentiment analysis in e-commerce platforms: a review of current techniques and future directions', IEEE Access, 11, pp. 90367–90382.

Ji, P., Zhang, H.-y. and Wang, J.-q. (2018) 'A fuzzy decision support model with sentiment analysis for items comparison in e-commerce', IEEE Transactions on Systems, Man, and Cybernetics: Systems, 49(10), pp. 1993–2004.

Kwon, H.J., Ban, H.J., Jun, J.K. and Kim, H.S. (2021) 'Topic modeling and sentiment analysis of online review for airlines', Information, 12(2), p. 78.

Munna, M.H., Rifat, M.R.I. and Badrudduza, A.S.M. (2020) 'Sentiment analysis and product review classification in e-commerce platform', in Proceedings of the 23rd International Conference on Computer and Information Technology (ICCIT). IEEE, pp. 1–6.

Poria, S., Cambria, E., Bajpai, R. and Hussain, A. (2017) 'A review of affective computing: from unimodal analysis to multimodal fusion', Information Fusion, 37, pp. 98–125.

Roelleke, T. and Wang, J. (2008) 'TF-IDF uncovered: a study of theories and probabilities', in Proceedings of the 31st Annual International ACM SIGIR Conference. ACM, pp. 435–442.

Sanh, V., Debut, L., Chaumond, J. and Wolf, T. (2019) 'DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter', arXiv preprint arXiv:1910.01108.

Singh, U., Saraswat, A., Azad, H.K., Abhishek, K. and Shitharth, S. (2022) 'Towards improving ecommerce customer review analysis for sentiment detection', Scientific Reports, 12(1), p. 21983.

Taboada, M., Brooke, J., Tofiloski, M., Voll, K. and Stede, M. (2011) 'Lexicon-based methods for sentiment analysis', Computational Linguistics, 37(2), pp. 267–307.

Vanaja, S. and Belwal, M. (2018) 'Aspect-level sentiment analysis on e-commerce data', in Proceedings of the 2018 International Conference on Inventive Research in Computing Applications (ICIRCA). IEEE, pp. 1275–1279.

Yang, L., Li, Y., Wang, J. and Sherratt, R.S. (2020) 'Sentiment analysis for e-commerce product reviews in Chinese based on sentiment lexicon and deep learning', IEEE Access, 8, pp. 23522–23530.